

Jose E. Rodríguez-Ríos

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joserico00.github.io

Ph.D. student and computer scientist with national-laboratory experience in cybersecurity, AI/HPC, and research software.

EDUCATION

Ph.D., Software Engineering , University of Puerto Rico - Mayagüez	Aug 2026 - Present
M.S., Computer Science , University of Puerto Rico - Río Piedras GPA: 4.0/4.0	Aug 2022 - May 2025
B.S., Computer Science , University of Puerto Rico - Río Piedras Minor in Cybersecurity	Aug 2015 - Dec 2020

TECHNICAL SKILLS

Programming languages: Python, C, C++, Assembly

AI and HPC: PyTorch, Hugging Face, TensorFlow, transformers, SLURM, GPU computing, Jupyter

Cybersecurity: Vulnerability assessment, reverse engineering, Nmap, OpenVAS, Ghidra, Wireshark, Scapy

Systems and DevOps: Linux, Docker, GitLab CI/CD, Git, Elasticsearch, Flask, Streamlit, SQLite, MySQL, OpenVPN

EXPERIENCE

Research Intern, Argonne National Laboratory, Lemont, IL Jun 2026 - Aug 2026

- Built DNA sequence-classification workflows with PyTorch, Hugging Face, and DNABERT on Perlmutter, Polaris, and Aurora; compared 6-mer and BPE tokenization (81% vs. 80% accuracy; 195 vs. 34 tokens per sequence).
- Led a 10-student team as project lead: scoped a 10-12-hour project, wrote Jupyter/SLURM lessons and CPU-versus-GPU experiments, and coached final presentations; its fine-tuned DNABERT reached 91.7% evaluation accuracy on CTCF data.

Volunteer Software Developer, Abeyno Coop solar cooperative, Salinas, PR 2024 - Present

- Built Solar Fleet Reporter, a 1,700-line Streamlit app with 88 automated checks that pulls energy data from Sol-Ark, FranklinWH, and sonnen portals, applies net-metering billing rules, and exports CSV/Excel reports and printable homeowner statements.

DevOps Intern, Lawrence Livermore National Laboratory, Livermore, CA Jun 2024 - Aug 2024

- Built a containerized Elasticsearch store for Ghidra BSim binary features so researchers can query malware-like samples at scale, with GitLab CI/CD for deployment.
- Deployed a Docker-hosted Ollama (Llama 3) service for the PaperQA PDF-reading application.

Graduate Research and Teaching Assistant, UPR Río Piedras, San Juan, PR Aug 2022 - May 2025

- Developed NVSRCO, an open-source vulnerability scanner using Python, Nmap, and OpenVAS, with CSV reporting, a Dash dashboard, and WireGuard links to remote networks.
- Led Operating Systems, Computer Architecture, and Cybersecurity labs (Linux, Ghidra, Wireshark, Scapy); authored Capture the Flag (CTF) challenges in reverse engineering, cryptography, web, and forensics for about 60 Puerto Rico high-school students.

Cybersecurity Intern, GM Sectech, San Juan, PR Jun 2022 - Aug 2022

- Built a Python/Twilio smishing (SMS phishing) simulator for awareness training; collaborated with penetration testers on credential-theft and domain-compromise scenarios; trained on Metasploit, Hashcat, and Burp Suite.

SELECTED PROJECTS *(full list at joserico00.github.io)*

Machine Learning for Security: Trained a TensorFlow CNN on the 9,401 image, 25-family Malimg dataset to 97.7% validation accuracy and built an IoT-23 traffic pipeline with capture-aware splits to prevent data leakage.

yt2ogg: Built a desktop and command-line audio tool that finds a track's beat grid with a dynamic-programming tracker in pure NumPy and renders gapless loops, packaged as standalone Windows executables with self-tests.

AWARDS AND ACTIVITIES *(4 of 7; full list in CV and joserico00.github.io)*

Peer Mentor, Intro to HPC Bootcamp, Argonne (2025): mentored 5 students designing HPC under a 12 MW cap.

BSides Puerto Rico CTF (2024): 1st place of 30 teams; reverse engineering, cryptography, exploitation.

Bloomberg HSI Tech Summit (2024): 1 of 29 selected from 400+ applicants; Summer Accelerator invitee.

Tracer Fire Blue Team, ACSAC (2019): 2nd place team of 40 participants; SIEM/IDS, Autopsy, Ghidra.